Gaia Pro VS + STP Multiplayer (Netick)

Technical Documentation

STP_Carryable_Log Falling Through Terrain After Harvesting Trees

Document Version: 1.0
Author: Steven Trautz (Community Contribution)
Status: Tested and Verified

---

1. Overview

This document describes a possible fix for an issue encountered while integrating **Gaia Pro VS** with **Survival Template Pro Multiplayer (Netick)**.

Within the project environment listed below, harvesting the default `STP_Harvestable_ConiferTree` caused the spawned `STP_Carryable_Log` objects to immediately fall through the terrain after spawning.

The implementation described in this document resolved the issue without requiring modifications to Gaia, terrain settings, physics, or the affected carryable prefabs.

Important

This is **not** an official XWizard Games fix.

It is a community-tested solution that resolved the issue in the environment described below.

---

2. Project Environment

Component	Version
Unity	6000.0.67f1
Render Pipeline	Universal Render Pipeline (URP)
Gaia Pro VS – Terrain, Trees, Grass & Water for Unity 6	4.2.4
Book of the Dead: Biome for Gaia Pro VS	4.2.3
Survival Template PRO	1.5.3
Survival Template PRO Multiplayer (Netick)	1.5.5.2
Emerald AI 2025	1.3.3
Emerald AI Multiplayer (Mirror / Netick)	1.3.3.2
Netick	1.1.21
Steamworks.NET	2025.163.0

3. Affected Objects

Harvestable Object

STP_Harvestable_ConiferTree

Spawned Object

STP_Carryable_Log

---

4. Issue Description

After cutting down an `STP_Harvestable_ConiferTree`, the carryable logs were spawned successfully.

Immediately after spawning, however, the logs continued falling through the terrain and eventually disappeared below the world.

As a result, the logs could no longer be collected by the player.

The issue was consistently reproducible throughout the project.

---

5. Expected Behavior

- Tree falls normally.
- Carryable logs spawn.
- Logs land on the terrain.
- Logs remain collectable.

---

6. Actual Behavior

- Tree falls normally.
- Carryable logs spawn.
- Logs ignore the terrain after spawning.
- Logs continue falling below the world.

---

7. Technical Analysis

During testing, the spawned object was consistently instantiated at the expected world position.

No issues were found with the spawn location, terrain collider, Rigidbody configuration, collision settings, or Gaia integration itself.

The issue was consistently resolved after explicitly initializing the `SimplePositionSyncer` network state immediately after spawning.

This suggests that the initial synchronized position and rotation did not match the object's actual spawn transform during the first synchronization step.

As a result, the spawned object received an incorrect synchronized state immediately after instantiation, causing it to continue falling below the terrain.

While additional investigation would be required to determine the exact synchronization sequence inside Netick, explicitly initializing the network state resolved the issue consistently within the tested project.

---

8. Implementation

Step 1 – Update `SimplePositionSyncer.cs`

File

`SimplePositionSyncer.cs`

Locate the " `public void SetActiveState(bool active)`" method inside the `SimplePositionSyncer` class.

Immediately below this method, insert the following code:

```
public void ForceState(Vector3 position, Quaternion rotation)
{
    if (Sandbox == null || !Sandbox.IsServer)
        return;

    NetworkedPosition = position;
    NetworkedRotation = rotation;
}
```

After the change, this section should look like this:

```
public void SetActiveState(bool active)

{

    Enabled = active;

    enabled = active;

}


public void ForceState(Vector3 position, Quaternion rotation)

{

    if (Sandbox == null || !Sandbox.IsServer)

        return;


    NetworkedPosition = position;

    NetworkedRotation = rotation;

}


public override void NetcodeIntoGameEngine()

{

    if (enabled)

    {

        transform.position = NetworkedPosition;

        transform.rotation = NetworkedRotation;

    }

}
```

Step 2 – Update `HarvestableFallBehaviour.cs`

File

`HarvestableFallBehaviour.cs`

At the top of the file, add the following namespace:

using XWizardGames.Integrations.FPSCore.Multiplayer;

Locate the following line inside the `SpawnItems(Transform resourceTransform)` method:

itemTransform.SetPositionAndRotation(itemPosition, itemRotation);

Immediately after this line, insert the following code:

```
if (itemInstance.TryGetComponent(out SimplePositionSyncer positionSyncer))
{
    positionSyncer.ForceState(itemPosition, itemRotation);
}
```

After the change, this section should look like this:

```
var itemInstance = sand.NetworkInstantiate(_itemPrefab.gameObject, itemPosition,
itemRotation);

var itemTransform = itemInstance.transform;



itemTransform.SetPositionAndRotation(itemPosition, itemRotation);



if (itemInstance.TryGetComponent(out SimplePositionSyncer positionSyncer))

{

    positionSyncer.ForceState(itemPosition, itemRotation);

}



itemInstance.GetComponent<Rigidbody>().AddForce(

    ItemForce * Mathf.Sign(Random.Range(-100f, 100f)) * itemTransform.right,

    ForceMode.Impulse);
```

---

9. Result

After applying both implementation steps:

- ✔ Carryable logs spawn correctly.
- ✔ Logs remain on the terrain.
- ✔ `SimplePositionSyncer` remains fully enabled.
- ✔ No Gaia modifications required.
- ✔ No terrain modifications required.
- ✔ No prefab modifications required.
- ✔ No collider modifications required.
- ✔ No Rigidbody modifications required.
- ✔ No physics settings changed.
- ✔ No spawn offset adjustments required.

The issue could no longer be reproduced in the tested environment.

---

10. Demonstration

A complete demonstration of this issue, the implementation steps, and the final result is available as a separate video.

The demonstration includes:

- Verification of the STP Multiplayer installation
- Project package versions
- Creating and registering a multiplayer scene using **Level Definition** and **MP Hook**
- Issue reproduction
- Debugging using `EditorApplication.isPaused`
- Implementation of both code changes
- Final verification after applying the fix

A YouTube link will be provided together with this documentation.

---

11. Notes

This fix has only been tested with the project configuration listed in this document.

It is shared as a community contribution in the hope that it may help other developers integrating **Gaia Pro VS** with **Survival Template Pro Multiplayer (Netick)**.

The fix is intentionally provided as source code snippets instead of complete files to simplify integration into projects that already contain custom modifications.